

Biological platforms for the production of antimicrobial peptides: Plants, insects, and mammalian cells systems

Luis H. Reyes^a and Juan C. Cruz^b

^aDepartment of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia, ^bDepartment of Biomedical Engineering, Universidad de los Andes, Bogotá, Colombia

Antimicrobial peptides (AMPs) not only exhibit a wide range of antimicrobial activities but also have immune-modulating and wound-healing properties. Their ability to interact with immune cells, cytokines, and chemokines makes them promising candidates for developing adjunctive treatments for infectious diseases and wound healing (Afacan et al., 2013; Lai & Gallo, 2009; Paulsen & Garreis, 2011). Ongoing research investigates the use of AMPs in combination with conventional antibiotics to improve their efficacy and reduce the risk of antibiotic resistance.

Recombinant DNA technology is a central tool in biotechnology and is essential for the therapeutic production of AMPs. This technology relies on the capacity to cut and manipulate DNA molecules to generate new recombinant DNA that can be introduced into living cells. The target DNA is extracted from its natural source or amplified through polymerase chain reaction (PCR) and then cloned into an appropriate vector to ensure stable and efficient expression in the host organism (Sambrook et al., 1989).

Chapter 8 will examine plant, insect, and mammalian cell systems as biological platforms for the production of AMPs (Fig. 8.1). We will discuss the benefits and limitations of each platform, as well as the obstacles that must be overcome to increase their efficiency and decrease production costs. Additionally, we will highlight the most promising AMPs produced using these platforms and their potential applications in the development of novel antimicrobial therapies and wound healing treatments.

1 Expression of antimicrobial peptides in plants

Antibiotic-resistant pathogens pose an ever-increasing threat, necessitating research into novel methods for combating these microorganisms. The production of AMPs in plants is one of these methods. This section of the chapter discusses the methods used to express these peptides, including heterologous and direct gene transfer techniques, as well as the plant expression systems developed for this purpose. Our objective is to provide a comprehensive understanding of the current state of the field and possible future directions for the development and application of plant-expressed AMPs by examining pertinent examples, benefits, and drawbacks of each method. Several AMPs have been produced in plants, and some are listed in Table 8.1.

1.1 Heterologous expression

This section will delve deeper into the heterologous expression of AMPs in plants, covering topics such as optimizing gene constructs, expression vectors, and promoters, as well as addressing challenges associated with protein processing and potential toxicity.

- *Optimizing gene constructs:* To successfully express AMPs in plants, it is crucial to optimize the gene constructs used for transformation. This can involve modifying the codon usage to better match the host plant's preferred codons, which can increase the efficiency of translation and lead to higher protein expression levels (Watts et al., 2021). Additionally, incorporating specific protein targeting sequences can ensure that the expressed AMP is directed to the appropriate

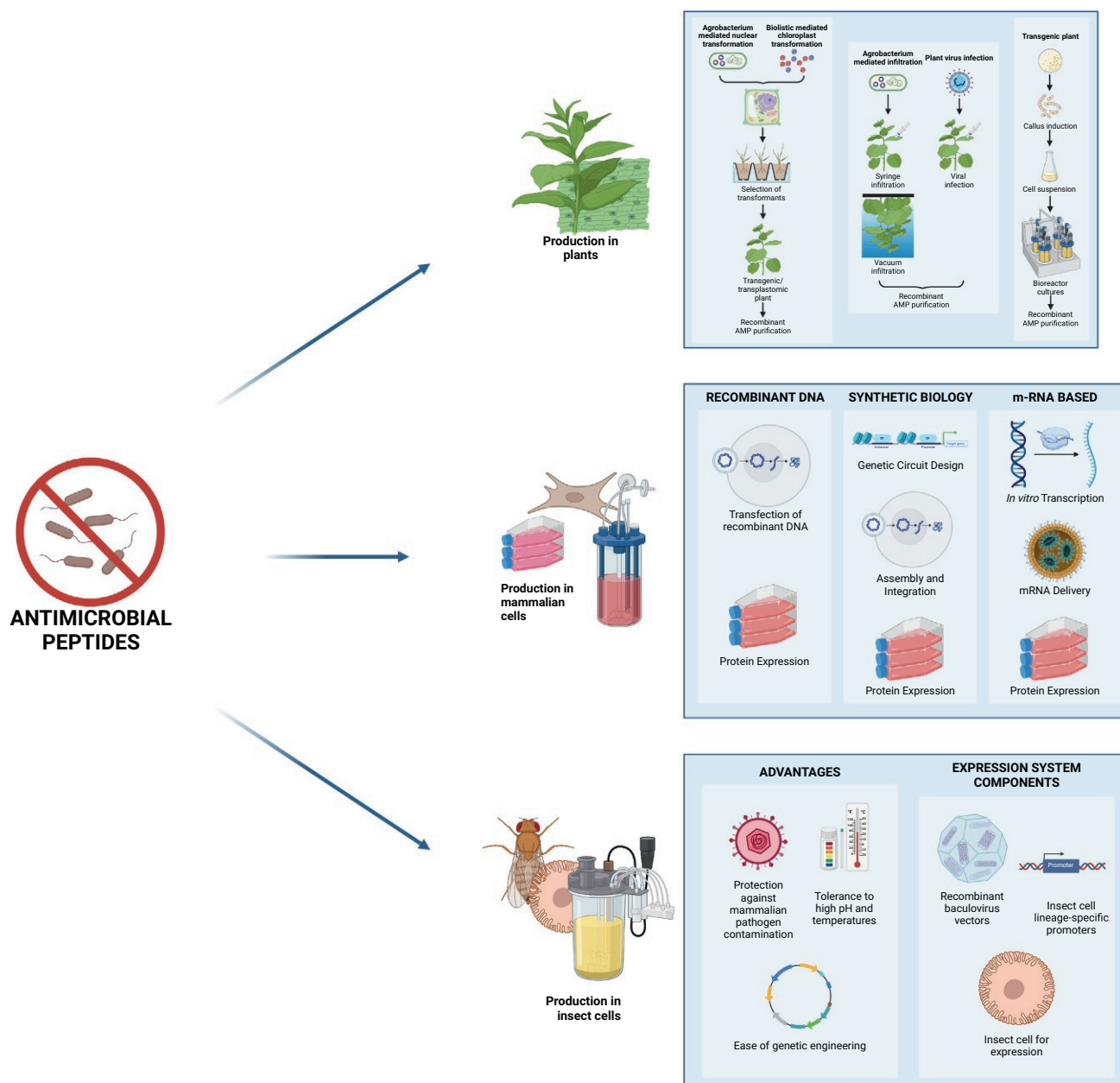


FIG. 8.1 Comparative strategies for antimicrobial peptide production in biological systems: From plants to insect cells approaches. This figure illustrates the diverse strategies for antimicrobial peptide production across various biological systems, including plants (A), mammalian cells (B), and insect cells (C).

subcellular compartment, such as the endoplasmic reticulum or vacuoles, where it can be properly processed and folded (Nosaki et al., 2021).

- **Expression vectors and promoters:** The choice of the expression vector and promoter can also significantly impact the efficiency and yield of AMP expression in plants. Expression vectors must be compatible with the chosen plant expression system and should allow for easy selection and identification of transformed plants. Commonly used plant expression vectors include binary vectors for *Agrobacterium*-mediated transformation and viral vectors for transient expression (Shah et al., 2013). Promoters, which control the transcription of the foreign gene, are another critical factor in heterologous expression. Strong constitutive promoters, such as the cauliflower mosaic virus (CaMV) 35S promoter, can drive high levels of gene expression in most plant tissues. However, in some cases, the use of tissue-specific or inducible promoters may be more suitable, particularly if the AMP is toxic to the host plant or if the goal is to target expression in specific tissues or developmental stages (Laxa, 2017).

TABLE 8.1 Antimicrobial peptides expressed in several hosts.

Name	Source	Host	Direct Expression?	Masked?	Expression	Carrier protein	Cleavage site	Yield (mg/L)	References
AP24 osmotine	<i>Nicotiana tabacum</i>	Potato	Y	N	Constitutive	–	–	–	Rivero et al. (2012)
C4V3	Synthetic	<i>N. tabacum</i>	Y	N	Constitutive	–	–	–	Rubio-Infante et al. (2012)
Cecropin B (CB)	<i>Hyalophora cecropia</i> (giant silk moth)	Tomato	Y	N	Constitutive	–	–	–	Jan et al. (2010)
Cecropin P1	Sus spp.	Potato	Y	N	Constitutive	–	–	–	Zakharchenko et al. (2007)
Chicken alpha interferon (ChIFN-alpha)	<i>Gallus gallus</i>	Lettuce	Y	N	Constitutive	–	–	–	Song et al. (2008)
Defensin 1 (BrD1)	<i>Brassica rapa</i>	Rice	Y	N	Constitutive	–	–	–	Choi et al. (2009)
EtMIC2	<i>Eimeria tenella</i>	<i>N. tabacum</i>	Y	N	Constitutive	–	–	–	Sathish et al. (2011)
Hevein-like peptides pro-SmAMP1 and pro-SmAMP2	<i>Stellaria media</i>	<i>Arabidopsis</i>	Y	N	Constitutive	–	–	–	Shukurov et al. (2012)
Hordothionin (hth)	<i>Hordeum vulgare</i>	Apple	Y	N	Constitutive	–	–	–	Krens et al. (2011)
IFN SasalFN-alpha1	<i>Salmo salar</i>	Rice and potato	Y	N	Constitutive	–	–	–	Fukuzawa et al. (2010)
Metchnikowin (Mtk)	<i>Drosophila melanogaster</i>	Barley	Y	N	Constitutive	–	–	–	Rahnamaeian and Vilcinskis (2012)
MsDef1 defensin	<i>Medicago sativa</i>	Tomato	Y	N	Constitutive	–	–	–	Abdallah et al. (2010)
MsrA2 (N-methionine-dermaseptin B1)	<i>Phyllomedusa sauvagei</i> and <i>Phyllomedusa bicolor</i>	<i>N. tabacum</i>	Y	N	Constitutive	–	–	–	Yevtushenko and Misra (2007)
NmDEF02	<i>Nicotiana megalosiphon</i>	<i>N. tabacum</i>	Y	N	Constitutive	–	–	–	Portieles et al. (2010)

Continued

TABLE 8.1 Antimicrobial peptides expressed in several hosts—cont'd

Name	Source	Host	Direct Expression?	Masked?	Expression	Carrier protein	Cleavage site	Yield (mg/L)	References
NmDEF02	<i>Nicotiana megalosiphon</i>	Potato	Y	N	Constitutive	—	—	—	Portieles et al. (2010)
Pal and Cpl-1	Phages infecting <i>Streptococcus pneumoniae</i>	<i>N. tabacum</i>	Y	N	Constitutive	—	—	—	Oey et al. (2009)
Penaeidin4-1	<i>Litopenaeus setiferus</i>	<i>Agrostis stolonifera</i> (creeping bentgrass)	Y	N	Constitutive	—	—	—	Zhou et al. (2011)
Retrocyclin-101 and Protegrin-1	Artificial AMP based on rhesus monkey (<i>Macaca mulatta</i>) circular minidefensins	<i>N. tabacum</i>	Y	N	Constitutive	—	—	—	Lee et al. (2011)
RsAFP2	<i>Raphanus sativus</i>	Wheat	Y	N	Constitutive	—	—	—	Zhao Li et al. (2011)
RsAFP2	<i>R. sativus</i>	Rice	Y	N	Constitutive	—	—	—	Jha and Chattoo (2010)
Thanatin	<i>Podisus maculiventris</i>	Rice	Y	N	Constitutive	—	—	—	Imamura et al. (2010)
Thionin (aHT)	<i>H. vulgare</i>	Sweet potato	Y	N	Constitutive	—	—	—	Muramoto et al. (2012)
Trichokonins	<i>Trichoderma pseudokoningii</i>	<i>N. tabacum</i>	Y	N	Constitutive	—	—	—	Luo et al. (2010)
Wasabi defensin	<i>Wasabia japonica</i>	"Egusi" melon	Y	N	Constitutive	—	—	—	Ntui et al. (2010)

- *Protein processing and folding*: One of the challenges associated with the heterologous expression of AMPs in plants is ensuring proper folding and posttranslational modifications. These processes are critical for the function and stability of the expressed peptide. Some AMPs may require specific disulfide bond formation or glycosylation patterns, which can be challenging to achieve in a heterologous expression system. To overcome this issue, researchers may need to coexpress molecular chaperones or modify the peptide sequence to facilitate proper folding and processing (Habibi et al., 2017).
- *Potential toxicity and host plant effects*: The expression of AMPs in plants can sometimes result in toxic effects on the host plant, potentially leading to growth defects or reduced fitness. To minimize these effects, researchers can explore various strategies, such as the use of tissue-specific or inducible promoters to limit AMP expression to certain tissues or developmental stages or the incorporation of protein targeting sequences to direct the AMP to specific subcellular compartments where it is less likely to cause toxicity (Hoelscher et al., 2022). Additionally, screening for compatible host plants that can tolerate the expression of the AMP without significant negative effects is another approach to address this challenge.

Examples of AMPs expressed heterologously in plants include:

1. *Defensins*: Plant defensins are small, cysteine-rich peptides with antimicrobial activity. Defensins from various sources, such as radish (Rs-AFP2) and pea (DRR230-c, DRR230-a), have been successfully expressed in tomatoes (Kostov et al., 2009) and tobacco (Lai et al., 2002) transgenic plants, respectively.
2. *Cecropins*: Cecropins are cationic AMPs originally isolated from insects. The expression of AMPs cecropin P1 (Zakharchenko et al., 2021) and cecropin A (Bundó et al., 2014) in *Brassica napus* L. and transgenic rice plants has been reported.
3. *Dermaseptins*: Dermaseptins are a group of naturally occurring AMPs found in the skin secretions of certain tree frogs. These peptides exhibit broad-spectrum antimicrobial activity against various pathogens, including fungi, oomycetes, bacteria, and even some viruses. Due to their potent activity and low toxicity to host cells, dermaseptins have emerged as a promising candidate for boosting plant resistance to a wide variety of pathogens. In a recent study, scientists fused the dermaseptin B1 (DrsB1) DNA coding sequence with the chitin-binding domain (CBD) of an effector protein from *Cladosporium fulvum* (Khademi et al., 2020).

Heterologous expression of AMPs in plants has the potential to produce novel antimicrobial agents at a low cost. Successful implementation of this strategy requires careful optimization of gene constructs, expression vectors, and promoters, as well as addressing challenges associated with protein processing, folding, and potential plant toxicity. Researchers can harness the power of plant-based heterologous expression systems for the development of effective and sustainable antimicrobial therapies by considering these factors and employing the appropriate strategies.

1.2 Direct gene transfer

Alternative strategies exist for achieving stable expression of AMPs in plants through direct gene transfer. Bypassing the need for a bacterial vector, these techniques involve the direct delivery of foreign DNA into plant cells. This section will provide an in-depth analysis of the various direct gene transfer methods, including electroporation, biolistic transformation (particle bombardment), polyethylene glycol (PEG)-mediated transformation, and silicon carbide fiber-mediated transformation, and discuss the challenges and benefits of each technique.

- *Electroporation*: Electroporation is a method that involves the application of an electric field to introduce foreign DNA into plant cells. This technique has been successfully used to generate transgenic plants expressing AMPs, such as defensins and cecropins (Chahardoli et al., 2018). Electroporation offers a rapid and straightforward method for the direct transfer of DNA into plant cells. It can be applied to a wide range of plant species and tissue types, offering flexibility in the choice of host plants for expression. However, electroporation is generally less efficient than *Agrobacterium*-mediated transformation and may be limited by factors such as the size and complexity of the DNA molecule, the plant species, and the tissue type used for transformation. Additionally, optimization of electroporation conditions is often required for each plant species to achieve high transformation efficiency (Ozyigit, 2020). An example of the use of electroporation was the expression of a lactoferrin chimera AMP in tobacco plants recombinant expression of LFchimera AMP in a plant-based expression system and its antimicrobial activity against clinical and phytopathogenic bacteria (Chahardoli et al., 2018).
- *Biolistic transformation (particle bombardment)*: Biolistic transformation, or particle bombardment, involves the use of a gene gun to deliver foreign DNA coated onto microscopic gold or tungsten particles into plant cells. This method has

been successfully employed for the production of transgenic plants expressing AMPs, such as defensins and cecropins (Hoelscher et al., 2022). Biolistic transformation offers several advantages, including the ability to deliver large DNA molecules and the potential for simultaneous cotransformation with multiple genes. This method is also compatible with a wide range of plant species and tissue types, providing flexibility in the choice of host plants for expression. Like electroporation, biolistic transformation is generally less efficient than *Agrobacterium*-mediated transformation and may result in complex integration patterns of the foreign gene, which can complicate subsequent characterization and analysis of the transgenic plants. Moreover, this method can be technically challenging and requires specialized equipment, such as a gene gun, for the delivery of DNA-coated particles into plant cells (Matsumoto & Gonsalves, 2012). To combat pathogenic infections, scientists have recently used biolistic transformation to engineer the marine diatom *Phaeodactylum tricornutum* to express the AMPs S-thanatin and bovine lactoferricin (bLFcin) (Baiden et al., 2023).

- **Microinjection:** Microinjection is the direct injection of foreign DNA into plant cells using a fine glass needle or micro-pipette. Transgenic plants expressing various proteins and peptides have been successfully produced using this method. Microinjection, on the other hand, is typically labor-intensive, time-consuming, and technically difficult, making it unsuitable for large-scale transgenic plant production. Microinjection's main advantage is that it eliminates the need for a bacterial vector by allowing for the direct introduction of foreign DNA into plant cells. This reduces the possibility of unintended gene transfer or the introduction of undesirable sequences, such as antibiotic-resistance genes (Klein et al., 1988). Despite its advantages, microinjection is a difficult technique that necessitates the use of specialized equipment and knowledge. In addition, the technique is constrained by the availability of suitable plant species and tissue types, and its low efficiency makes it less desirable for large-scale production of transgenic plants expressing AMPs (León & Fernández, 2007).
- **Polyethylene glycol (PEG)-mediated transformation:** Another direct gene transfer method is the use of polyethylene glycol (PEG) to facilitate the uptake of foreign DNA into plant cells, such as protoplasts or suspension-cultured cells. PEG-mediated transformation is a straightforward, economical, and rapid method for delivering foreign DNA into plant cells. This technique has been utilized successfully to generate transgenic plants expressing various proteins (Yang et al., 2022). PEG-mediated transformation can be less effective than other direct gene transfer techniques, such as electroporation and biolistic transformation. In addition, PEG-mediated transformation is primarily applicable to the transformation of protoplasts, which requires the enzymatic removal of the plant cell wall and can restrict the range of plant tissues that can be transformed (Kofer et al., 1998).

1.2.1 Optimization strategies for direct gene transfer methods

Alternative strategies exist for achieving stable expression of AMPs in plants through direct gene transfer. Researchers can employ a variety of optimization strategies to enhance the efficacy and success of these methods. This section will provide a comprehensive analysis of these strategies, with a focus on the optimization of gene constructs and vectors, the selection of suitable plant species and tissues, and the implementation of advanced gene delivery techniques.

The design of gene constructs and vectors can have a substantial effect on the efficiency of gene delivery and expression. Optimizing elements such as promoters, terminators, and regulatory sequences can increase the expression of the desired AMP.

- **Promoters:** The choice of promoter for driving gene expression is crucial. Strong promoters, such as the cauliflower mosaic virus (CaMV) 35S promoter or the maize ubiquitin (Ubi) promoter, can drive high levels of gene expression, resulting in an increase in AMP production (Pauli et al., 2004).
- **Terminators:** Terminators are the sequences that indicate the conclusion of transcription. The choice of terminator can influence the stability of mRNA transcripts and the levels of gene expression. Researchers can optimize the terminator sequence to ensure efficient transcription termination and improve mRNA stability (Wang, Kumar, et al., 2020).
- **Regulatory sequences:** Enhancers, silencers, and introns are all examples of regulatory sequences that can modulate gene expression. Researchers can manipulate gene expression to achieve specific results by adjusting the optimal sequences (To et al., 2021).

1.2.2 Selection of appropriate plant species and tissues

The efficiency of direct gene transfer techniques can be affected by factors such as plant species and tissue of choice. The ability to regenerate plant tissue, the target expression levels, and the method's compatibility with the plant species are all important considerations for researchers (Paszkowski et al., 1984).

- *Plant species compatibility*: Compatibility with direct gene transfer techniques may vary across plant species. When choosing a plant species for expression, researchers should think about how well genes can be delivered and whether or not transformed plants can be successfully regrown.
- *Tissue regeneration capacity*: For direct gene transfer methods to be effective, plant tissues must be able to regenerate into whole plants after receiving genes. Certain plant tissues are better suited for transformation due to their greater regenerative gene potential. These include immature embryos, calluses, and leaf explants.
- *Desired expression levels*: The expression levels can also be affected by the plant species and tissue types that are used. When deciding on a host for AMP production, researchers should take into account the innate expression capacities of various plant species and tissues.

To express proteins and peptides in plants, [Table 8.2](#) provides a comparison of heterologous expression and direct gene transfer.

1.3 Plant expression systems

1.3.1 Viral-based expression systems

The use of plant viruses as vectors for the expression of foreign genes in plants is the basis of viral-based expression systems. These systems are ideal for the production of AMPs in plants due to their rapid and high-level expression of recombinant proteins. Some common examples include the tobacco mosaic virus (TMV) system and the potato virus X (PVX) system.

- *Tobacco mosaic virus (TMV) system*: To express foreign genes, like those encoding AMPs, in infected plant cells, the TMV system uses a modified version of the TMV RNA genome. An appealing choice for AMP expression is the TMV system, which is well-known for its capacity to rapidly produce high levels of recombinant proteins ([Abrahamian et al., 2020](#)). One drawback is that this system is best used for temporary expression because the foreign gene is not permanently inserted into the plant genome. A tobacco mosaic virus (TMV)-based vector was used in a recent study to induce the expression of antifungal proteins (AFPs) in plant cells. The antifungal protein AfpB was produced in *Nicotiana benthamiana* plants using the TMV vector system ([Shi et al., 2019](#)).
- *Potato virus X (PVX) system*: Another viral-based expression system, the PVX system, employs a modified PVX genome to induce the expression of foreign genes in plants. The PVX system, like the TMV system, permits the high-level transient expression of recombinant proteins, such as AMPs ([Lico et al., 2015](#)). The versatility of the PVX system in infecting different plant hosts can be useful in finding the best plant species for AMP production.

TABLE 8.2 Comparison of heterologous expression and direct gene transfer in plants.

Method	Advantages	Tradeoffs	Challenges
Heterologous expression	• High level of protein expression • Ability to express complex proteins and peptides • Scalable production • Simplified downstream processing and purification	• Possible limitations in expression due to codon usage biases • Need for optimization of expression conditions • Potential decrease in host organism fitness • Dependency on host organism viability	• Limited range of host organisms for expression • Potential for incorrect post-translational modifications • Intellectual property and regulatory issues • High cost and technical expertise required
Direct gene transfer	• Stable integration of gene into the plant genome • Possibility to express the protein of interest in a specific tissue or developmental stage • No need for a host organism • Potential to express novel proteins not found in nature	• Random integration of the transgene • Potential silencing of transgene expression • Unpredictable levels of protein expression • Potential for the transfer of undesirable traits	• Low transformation efficiency • Time-consuming and labor-intensive • Intellectual property and regulatory issues • Difficulty in optimizing expression conditions and protein yield in plants

However, similar to the TMV system, the PVX system is limited by its transient nature and potential scalability issues due to the need for large-scale plant cultivation. A recent study showed the feasibility of the transient expression of the bifunctional AMP BP178 in *Nicotiana benthamiana* plants using the potato virus X (PVX) vector system (Baró et al., 2022).

1.3.2 *Agrobacterium*-mediated transformation

Stable expression of foreign genes in plants through *agrobacterium*-mediated transformation is a common practice. *Agrobacterium tumefaciens*, a bacterium found in soil, is the basis of this system due to its inherent capacity to transfer a segment of its DNA (T-DNA) into plant cells. A gene of interest, such as one encoding an AMP, can be introduced into the plant genome by inserting it into the T-DNA region of a binary vector and then using *Agrobacterium* to mediate the transformation (Wen et al., 2022). Since the foreign gene is permanently incorporated into the plant genome through *agrobacterium*-mediated transformation, it allows for the possibility of continuous AMP production in plants. This strategy is adaptable because it can be used with numerous plant species, making AMP expression much simpler. This system requires the generation of stable transgenic plants and subsequent characterization to ensure proper expression and functionality of the AMP, which can be more time-consuming and labor-intensive than transient expression systems.

The use of robust promoters, codon optimization, and protein targeting sequences are examples of optimization strategies for *Agrobacterium*-mediated transformation. To further enhance the efficiency of gene transfer and stable integration, scientists can make use of cutting-edge techniques for *Agrobacterium*-mediated transformation, such as floral dip transformation or *in planta* transformation (Diaz et al., 2016). To combat the widespread plant pathogen *Xanthomonas campestris* pv. *vesicatoria*, researchers have recently looked into an inducible expression of a de novo-designed AMP, SP1-1, in tomatoes (Diaz et al., 2016). The goals of the study were to find solutions to the problems that arise when AMPs are expressed in plants, such as the AMP's potential toxicity to the plant and its instability when exposed to the host's proteases. The use of an inducible expression system allows the plant to produce AMPs only when necessary, reducing the likelihood of any undesired side effects. Tomato plants were able to express the AMP functionally thanks to the use of a synthetic promoter derived from parsley (4XW2/4XS). Tomatoes were shown to be resistant to *X. campestris* pv. *vesicatoria* through the inducible expression of SP1-1. This method may increase crop yields and decrease the risk of drug resistance, making it a promising alternative for long-term plant protection against pathogens. The successful implementation of this strategy, however, depends on addressing potential tradeoffs such as phytotoxicity and stability. To improve global food security, more work is needed to optimize the expression of AMPs in other crops and against a wider variety of pathogens. This research highlights the feasibility of using inducible expression systems in combination with synthetic promoters to express AMPs in plants, providing a long-term strategy for the control of plant diseases.

1.3.3 *Plastid* transformation

Plasmid transformation, also known as chloroplast transformation, allows for the stable expression of foreign genes by inserting them into the plastid genome rather than the nuclear genome. Due to the maternal inheritance of plastids in most plant species, this method reduces the risk of horizontal gene transfer (HGT) and allows for high levels of protein expression, multigene engineering, and other benefits (Diaz et al., 2016). Dermaseptins and cathelicidins are just two examples of AMPs that have been successfully expressed in plants through plastid transformation. Unfortunately, this technique is currently only applicable to a select few plant species, primarily *Nicotiana tabacum* (tobacco), which can limit the availability of potential expression host plants.

To increase astaxanthin production beyond its current limits, scientists have recently demonstrated the successful genetic engineering of chloroplasts in *Haematococcus pluvialis*, a unicellular green alga (Wang, Cui, et al., 2020). Although it has been shown to have positive effects on human health, the production of astaxanthin is hampered by a number of factors. The study's primary goals were the successful expression of the AMP piscidin-4 and the expression of key enzymes for astaxanthin synthesis. The biolaphos resistance gene (*bar*) was used as a selectable marker to develop plastid engineering in *H. pluvialis* by creating a chloroplast transformation vector. Chloroplast transformation was performed using the biolistic method, a reliable strategy for *H. pluvialis* transformations. *H. pluvialis*'s growth rate in a liquid MCM medium was determined, and the introduced *antI* gene's protein product was tested using western blotting. Results showed that piscidin-4, an AMP, was successfully expressed in *H. pluvialis*. Piscidin-4 is one example of an AMP that has shown promise as an alternative to conventional antibiotics. The results raise the possibility of employing *H. pluvialis* as a platform for the expression of recombinant proteins. Progress in chloroplast genetic engineering in *H. pluvialis* paves the way for further study, which may one day result in the increased efficiency with which valuable compounds such as astaxanthin can be produced for use in human health.

1.4 Factors impacting plant expression systems

The effectiveness and versatility of plant expression systems for the generation of AMPs can be affected by a number of factors. The ability to achieve correct protein processing and folding, as well as the selection of a suitable host plant, is among three such factors. When choosing a plant expression system for the production of AMPs, researchers should give careful consideration to these factors and may employ optimization strategies to increase the efficiency, yield, and quality of the expressed peptides.

- *Choice of host plant:* The success of plant expression systems is highly dependent on the selection of the host plant. Host plants should be amenable to efficient gene delivery, protein expression, and processing using the selected expression system. Tobacco (*N. tabacum*), lettuce (*Lactuca sativa*), and rice (*Oryza sativa*) are just a few of the plant species that have served as hosts for the expression of proteins and AMP.
- *Agronomic traits:* Agronomic traits such as growth rate, biomass production, and cultivar suitability should all be taken into account when choosing a host plant. High-yield recombinant protein production can be achieved by using host plants with high rates of growth and biomass production while lowering production costs and simplifying the process by selecting plants that are easy to cultivate.
- *Protein expression and processing:* Host plants should also be selected based on their ability to express and process the desired AMPs efficiently. The expression of some AMPs may be better suited to certain plant species than others due to differences in protein expression levels or protein processing abilities.
- *Biosafety considerations:* Biosafety considerations should also be taken into account when selecting host plants. When selecting host plants for AMP expression, it is important to select species that are distantly related to native or agriculturally important species in the area where the plants are grown.
- *Efficiency of gene delivery:* The effectiveness of gene delivery is also crucial to the achievement of plant expression systems. The chosen expression system must be compatible with the host plant and allow for the efficient transfer of the foreign gene into the plant's genome or cellular machinery. Optimizing gene constructs, selecting suitable expression vectors, and employing cutting-edge gene delivery techniques such as electroporation or biolistic transformation are just a few of the methods that can be used to increase the success rate of introducing foreign genes into a plant's DNA.
- *Protein processing and folding:* The functionality and stability of the expressed AMPs depend on the proteins being processed and folded correctly. Heterologous expression systems can make it difficult to produce AMPs that have complex disulfide bond or glycosylation requirements. To overcome these difficulties, scientists have developed a number of solutions, such as the coexpression of molecular chaperones and alterations to the peptide sequence.

Further research and advances in plant expression systems, molecular techniques, and protein processing technologies are expected to improve the efficiency and scalability of AMP production in plants. This will enable the development of a diverse array of AMPs with potential applications in agriculture, medicine, and biotechnology.

In sum, the expression of AMPs in plants represents a novel and encouraging approach to the problem of rising antibiotic resistance. Researchers have been able to successfully produce AMPs with a variety of antimicrobial properties through various approaches, including heterologous and direct gene transfer methods and different plant expression systems. These plant-based expression systems offer several benefits over the more common microbial or animal-based expression platforms, including lower costs, greater scalability, and less environmental impact. While there is potential for optimizing AMP production using any of these approaches, each comes with its own set of difficulties and restrictions. Protein folding, posttranslational modifications, and undesired plant host responses are just a few of the challenges that have been identified in heterologous expression methods. However, there are potential risks associated with the release of genetically modified organisms into the environment, as well as regulatory and ethical concerns, which may arise from direct gene transfer.

Improving the expression and functionality of plant-derived AMPs will require further development of these methods and the investigation of alternative approaches. Integrating cutting-edge gene-editing tools such as CRISPR-Cas9 is a step in the right direction because it promises to improve the accuracy and efficiency of plant genetic modifications. The potential of this area may be further enhanced by the creation of novel plant expression systems that increase the stability, bioavailability, and activity of the expressed peptides. As we work toward a time when more potent antimicrobial treatments are available, it is clear that research into and practical application of AMPs expressed in plants will have a significant impact on the future of infection management and disease prevention. By removing roadblocks and realizing this novel approach's full potential, we may be able to create novel therapeutic strategies that can effectively combat a wide variety of pathogenic microorganisms and aid in the worldwide fight against the spread of antibiotic resistance.

2 Insect cell-based expression systems for antimicrobial peptide production

The use of insect cells for AMP production has several advantages over other expression systems (Almasia et al., 2017; Käßer et al., 2022). First, insect cells provide better protection against contamination by mammalian pathogens, which can be an issue when producing therapeutic proteins. Second, insect cells exhibit better tolerance to high pH and temperatures, which are necessary for the efficient production of an ample variety of AMPs. Finally, insect cells can be easily genetically engineered to express AMPs, which makes them a good choice for producing complex AMPs (Li et al., 2010; Sampaio de Oliveira et al., 2020).

Over the past 20 years, significant progress has been made in the development of insect cell-based expression systems. New methods have been developed to improve the efficiency of insect cell expression systems, including the use of recombinant baculovirus vectors, insect cell lineage-specific promoters, and inducible expression systems. New ways have been found to make insect cell expression systems work better. These include using recombinant baculovirus vectors, insect cell lineage-specific promoters, and expression systems that can be turned on and off (Kost et al., 2005; Sari-Ak et al., 2023). The availability of baculovirus vectors and insect cell-lineage-specific promoters has permitted the creation of efficient expression systems for a variety of AMPs (Grose et al., 2021). In addition, the use of inducible expression systems has enabled the regulation of protein expression over time (McGuire et al., 2004). This might be useful for producing AMPs that are toxic to the host in high concentrations or when improperly folded (Fjell et al., 2012; Otvos Jr, 2002). Using inducer-inducible expression systems has also led to the development of novel insect cell-based expression systems that may be useful for boosting the yield and quality of AMPs (Chang et al., 2018; Tokumoto et al., 2020).

Numerous advantages exist for the expression of AMPs in insect cells. For instance, the large cell volume of insect cells makes them suitable for large-scale production. In addition, the short doubling times of insect cells make them ideal for producing proteins and peptides with a short half-life (Bernard et al., 1994; Douris et al., 2006; Käßer et al., 2022). Finally, insect cells can be induced to express proteins by adding inducers such as tetracycline or isopropyl β -D-1-thiogalactopyranoside (IPTG). This is particularly useful for producing proteins or peptides difficult to express at high levels without inducing cell death (Grose et al., 2021; Toth et al., 2014). These features of insect cells make them a viable choice for the large-scale production of proteins and peptides. In addition, insect cells are easily transfected with expression vectors and can be used to produce substantial amounts of recombinant proteins (Grose et al., 2021).

Recombinant baculovirus expression vectors (BEVs) are an effective method for expressing foreign genes in insect cells (Contreras-Gómez et al., 2014; Felberbaum, 2015; Martínez-Solís et al., 2019). The gene of interest is under the control of a viral promoter in these vectors. Through engineering, additional regulatory sequences, such as the 5' untranslated region and the polyadenylation signal, can be added to the vectors (Fath-Goodin et al., 2006; Luckow, 1993). Following the infection of cells with a recombinant virus, the gene of interest is replicated alongside the genome of the vector. The resulting double-stranded DNA is subsequently incorporated into the genome of the host cell.

The viral immediate early (IE) gene is a crucial component of the baculovirus-based insect cell expression system, which regulates the expression of foreign genes in insect cells. The IE1 protein encoded by the IE gene is the primary regulator of late gene expression in insect cells infected with baculovirus. The IE1 protein binds to the promoters of the late genes and activates their transcription, resulting in the production of a broad spectrum of viral structural and non-structural proteins, as well as recombinant proteins of interest. To express foreign genes in insect cells using the baculovirus system, the foreign gene of interest must be transferred into the virus genome, which is then used to infect the insect cells (Chambers et al., 2018; Luckow & Summers, 1988). Typically, the foreign gene is inserted into a transfer vector containing a strong promoter, such as the polyhedrin promoter, which drives high levels of foreign gene expression in infected insect cells. Due to its high level of expression and ease of use, the polyhedrin promoter is one of the most widely used promoters in insect cell expression systems. Once the virus infects the insect cells, the IE1 protein is expressed, which then activates the expression of the late genes. In addition to the recombinant protein of interest, the late genes are responsible for the production of the viral structural proteins and enzymes required for virus replication.

The expression of the late genes is tightly controlled by the IE1 protein, which regulates the expression of mRNA capping and export factors, ribosome-binding sites, and other factors required for efficient protein expression in insect cells. Other viral proteins, such as the p35 protein, can also influence foreign gene expression in insect cells, in addition to the IE1 protein. Caspases are proteases that play a crucial role in apoptosis or programmed cell death; p35 is an inhibitor of caspases. The p35 protein inhibits caspase activation, which can increase cell survival and promote the expression of foreign genes in insect cells that have been infected. However, the p35 protein can also inhibit the immune response of insect cells, thereby reducing viral replication and foreign gene expression (Miller, 1988; Vialard et al., 1995).

Using the baculovirus system, the efficiency of foreign gene expression in insect cells can be influenced by a number of factors. One critical factor is the choice of the promoter used to drive the expression of the foreign gene. Different promoters

have varying strengths, and the choice of the promoter will depend on the level of expression required for the recombinant protein of interest. In addition to the choice of the host cell, the multiplicity of infection (MOI), the timing of induction, and the use of inducible promoters, other variables can affect the expression efficiency. Overall, the baculovirus-based insect cell expression system is a powerful tool for the expression of foreign genes in insect cells, and the expression of the viral IE1 gene plays a critical role in regulating the expression of late genes and the production of recombinant proteins of interest (Blissard & Rohrmann, 1990; Contreras-Gómez et al., 2014; Hitchman et al., 2009).

Some of the most important advantages and disadvantages of BEVs are shown as follows (Chen et al., 2011; Kidd & Emery, 1993; Kost & Kemp, 2016; Trowitzsch et al., 2010):

- They exhibit high expression levels and are suited for the mass production of recombinant proteins or peptides.
- They can be utilized to overexpress proteins and peptides with varying solubilities.
- They lack endogenous genes and are compatible with plasmid-free cell lines (e.g., S2 cells).
- They can be propagated in insect cells to high titers.
- They are simple to employ and have a high rate of success in producing recombinant proteins and peptides.
- Their size may render them unsuitable for certain applications. In addition, long DNA molecules may stimulate the immune response of mammalian cells, resulting in the formation of antibodies against the viral capsid protein.
- They are stable but may vary from batch to batch. Therefore, it may be difficult to identify clones that produce high yields of particular proteins or peptides.
- They are difficult to sequence because DNA cleavage in insect cells produces multiple fragments.
- They require large-scale, time-consuming, and costly production.
- They may exhibit host-specific restriction factors that limit their ability to spread.

Recently, a new expression system called MultiBac has been made available for the expression of multiprotein complexes. MultiBac is a modified baculovirus that has been engineered. Several enhancements and modifications have been implemented to facilitate the use of MultiBac, including the incorporation of ExpiFectamine Sf, a next-generation cationic lipid-based transfection reagent. CRISPR/Cas9 technology has emerged as a useful tool not only for gene editing applications but also for the use of synthetic transcriptional regulators in baculovirus systems. CRISPRi involves the recruitment of chimeric dCas9, sgRNA, transcriptional effector complex, and RNA polymerase to gene sequences. The cargo capacity of CRISPR-based gene editing can surpass that of conventional viral vector systems (Sari-Ak et al., 2023).

Insect cells have successfully expressed cecropins, attacins, gloverins, moricins, and defensins, among other AMP families. Insects produce cecropins, a class of AMPs with broad-spectrum activity against bacteria, fungi, and viruses. Cecropin A has been expressed in insect cells using a baculovirus vector and a fusion construct containing the antibody-binding part of protein A (Andersons et al., 1991; Hellers et al., 1991). Attacins are a family of AMPs found in the fat body of *Hyalophora cecropia*, the giant silk moth. Attacins have been demonstrated to be antimicrobial against Gram-negative bacteria and fungi, and some of them have been expressed in insect cells (Gunne et al., 1990; Sun et al., 1991). Gloverins are another class of AMPs found in the hemolymph of *Bombyx mori* silkworms, with antibacterial and antifungal activity. A number of gloverins were successfully expressed in insect cells using BEVs (Kawaoka et al., 2008; Lundström et al., 2002). Moricins are a family of AMPs that are found in *B. mori* silkworms and possess remarkable antimicrobial activity against Gram-negative bacteria and fungi. Multiple moricins have been detected in insect cells (Furukawa et al., 1999). Insects produce defensins, a family of AMPs with broad-spectrum antimicrobial activity against bacteria, fungi, and viruses. Multiple defensins have been expressed in insect cells, indicating their antimicrobial potential (Aono et al., 2006; Corrales-Garcia et al., 2011).

Despite the advantages of this technology, several obstacles must be surmounted before commercial quantities of effective AMPs can be manufactured. The selection of an appropriate insect cell line for AMP production is one of the major obstacles. Several insect cell lines, such as Sf9 and Sf21, have been utilized for the production of recombinant proteins, but their suitability for AMP production varies depending on the specific peptide and the desired biological activity (Druzinec et al., 2013; Geisse et al., 1996). Another obstacle is the optimization of the AMP production expression system. Using powerful promoters, such as the polyhedrin promoter or the baculovirus immediate-early promoter, can increase the level of expression of recombinant AMPs (Grose et al., 2021). However, high-level protein expression can also result in the formation of inclusion bodies or improper protein folding, reducing their biological activity. To achieve the optimal yield and biological activity of recombinant AMPs, it is necessary to optimize the expression conditions, including the culture medium, temperature, and induction time (Gorda et al., 2021; Müller et al., 2015). Many AMPs rely on posttranslational modifications, such as glycosylation, for their biological activity, and insect cells are able to perform these modifications. However, the extent and pattern of glycosylation can vary depending on the insect cell line and culture conditions, which can impact the biological activity of the recombinant protein. To achieve the desired glycosylation pattern and biological

activity, it is essential to characterize the glycosylation pattern of recombinant AMPs produced in insect cells and optimize the culture conditions (Agathos, 2007; Druzinec et al., 2013; Müller et al., 2015; Sari et al., 2016).

The purification of the recombinant protein is a further obstacle to AMP production in insect cells. To facilitate their purification, AMPs are frequently synthesized as fusion proteins with tags such as His-tag or GST. However, these tags can affect the biological activity of the recombinant protein, and removing them can be difficult. To achieve a high level of purity and biological activity of the recombinant AMP, it is crucial to optimize the purification conditions, including the selection of the chromatography resin and the buffer composition (Baek et al., 2011; Harper & Speicher, 2011; Wibowo & Zhao, 2019). In addition to the difficulties associated with AMP production in insect cells, the characterization of the biological activity of recombinant AMPs presents obstacles. AMPs can have various modes of action, such as membrane disruption, intracellular targeting, and immunomodulation, and their biological activity can be affected by a number of variables, including the microorganism being targeted, the concentration of the peptide, and the presence of host defenses (Girard et al., 2008; Sampaio de Oliveira et al., 2020). Consequently, it is essential to perform a comprehensive characterization of the biological activity of recombinant AMPs, including their minimum inhibitory concentration (MIC), mode of action, and cytotoxicity to host cells.

3 Technologies for expressing antimicrobial peptides in mammalian cells

This section provides a thorough analysis of the key factors influencing the expression of AMPs in mammalian cells. This section explores the trade-offs involved in balancing various factors and the difficulties associated with various approaches, including recombinant DNA technology, synthetic biology, and mRNA-based expression. We examine the trade-offs involved in balancing various factors and the difficulties associated with these approaches.

3.1 Recombinant DNA technology

Recombinant DNA (rDNA) technology is a potent molecular biology instrument that enables gene manipulation and expression in living organisms. This technology has been extensively utilized in the field of AMP research to enable the expression of these peptides in diverse mammalian cell systems (Khan, 2013).

There are multiple methods, including viral and nonviral gene delivery systems, for introducing the gene encoding an AMP into mammalian cells. Viral systems, including adenoviruses, adeno-associated viruses, and lentiviruses, are extremely effective at delivering genes to mammalian cells (Nayerossadat et al., 2012). However, viral systems may be associated with safety concerns, including insertional mutagenesis, immune responses, and the possibility of viral recombination events (Bulcha et al., 2021). Nonviral gene delivery systems include electroporation, liposomes, and nanoparticles, which offer advantages in terms of safety, production simplicity, and the potential for targeted delivery (Bono et al., 2020). However, these systems typically have a lower transfection efficiency than viral systems, which can limit their applicability in certain situations.

The choice of the promoter, which regulates the level and timing of gene expression, is an additional factor to consider in the expression of AMPs. To achieve high levels of expression, potent promoters, such as the cytomegalovirus (CMV) promoter, are frequently employed. In certain circumstances, such as when investigating the physiological role of AMPs, it may be preferable to employ tissue-specific or inducible promoters (Ede et al., 2016).

Nonetheless, the expression of AMPs in mammalian cells presents a number of trade-offs and obstacles. The need for proper protein folding and posttranslational modifications must be weighed against the lower protein production rates of mammalian cells in comparison with bacterial and yeast cells. In addition, the choice of vector and promoter can influence the overall efficacy, biosafety, and scalability of the expression system.

The potential cytotoxicity of AMPs to the host mammalian cells represents a further obstacle. High levels of AMP expression can disrupt cellular processes and compromise cell viability, which can negatively impact the yield and quality of the desired product (Erdem Büyükkiraz & Kesmen, 2022). This issue can be mitigated by employing inducible promoters that permit controlled expression or by incorporating secretion signals to facilitate the release of AMPs from producing cells.

The following are examples of AMP expression in mammalian cells using recombinant DNA technology:

- CAMA-syn expression in pulmonary epithelium:** The expression of the AMP CAMA-syn in pulmonary epithelium has emerged as a potential method for combating intracellular bacterial infections, such as *Mycobacterium tuberculosis* (Mtb). CAMA-syn is an antibacterial and antitumor peptide hybrid derived from Cecropin A (CA) and magainin 2 (MA). Researchers expressed CAMA-syn in A549 human lung epithelial cells and mouse lung epithelial cells using

recombinant adenoviruses, effectively inhibiting the growth of intracellular bacteria such as BCG (*Bacillus Calmette-Guérin*). The expression of CAMA-syn is greatly affected by the choice of promoter and the way recombinant adenovirus is introduced into the body. SPC, a promoter that works only in lung epithelial cells, showed targeted gene expression in lung epithelial cells and reduced off-target effects. The tracheal infusion was found to be the most effective way to keep CAMA-syn expression stable in the lung while reducing immune activation. But there are some problems and trade-offs, such as the possibility of inflammation and immune response after tracheal infusion, which need to be carefully managed. Also, the fact that antibacterial activity is different between cell lines such as Phoenix and A549 cells shows that the expression system and promoter activity need to be tweaked more. CAMA-syn expression in the pulmonary epithelium is a promising strategy for combating intracellular bacterial infections. Further research is required to optimize the expression and delivery methods, balancing specificity, efficiency, and potential side effects (Yue et al., 2020).

- *Expression of thanatin in HEK293 cells:* The HEK293 (Human Embryonic Kidney) cell line was used to make recombinant thanatin, and its antimicrobial properties were tested against different human bacterial pathogens. GenScript Corporation ensured that the coding sequence for thanatin would function optimally in HEK293 cells before manufacturing it. For efficient secretion, an Ig Kappa chain from *Mus musculus* (Igk) was used as a secretion leader. For plasmid amplification, the thanatin sequence was cloned into a plasmid vector and transformed into *E. coli* strain DH5. The recombinant plasmid was then transfected into HEK293 cells, and the expression of thanatin was analyzed using Tricine SDS-PAGE. Minimal inhibitory concentration (MIC) tests were conducted on numerous bacterial strains, including Methicillin-resistant *Staphylococcus aureus* (MRSA), to evaluate antimicrobial activity. Using MTT assays on human cells, the cytotoxicity of thanatin at various concentrations was not found to be significant. This study demonstrates the capability of the HEK293 expression system to produce recombinant thanatin. Although *E. coli* BL21 is commonly used for protein expression, it is unsuitable for small, highly expressed AMPs such as thanatin. The successful expression of thanatin in HEK293 cells, its demonstrated antimicrobial properties, and its low cytotoxicity on mammalian cells make it a promising candidate for the treatment of human diseases caused by antibiotic-resistant bacteria. Future research should investigate the synergy of thanatin with other components, such as antibiotics, essential oils, and other AMPs, to increase its therapeutic potential (Tanhaeian et al., 2020).
- *Expression of carcinoembryonic-antigen-human-b-defensin-2 in fibroblasts:* Human beta-defensin-2 (HBD-2) is a potent AMP against Gram-negative bacteria. In tumor cells, it was expressed and fused with the gene for carcinoembryonic antigen (CEA-HBD-2). HBD-2 is primarily secreted by epithelial cells of the respiratory tract mucosa, whereas CEA protein is a cancer protein found in intestinal tumor cells. A human colon cancer cell expression system with high HBD-2 expression was created by integrating the HBD-2 recombinant gene with the CEA gene and utilizing retroviral vectors for high expression. This system may allow human cells to produce AMPs in large quantities. To create a recombinant retroviral expression vector (PLNCX2-CEA-HBD-2), the full-length fragment of the CEA-HBD-2 gene was cloned using Primer Premier 5.0 software. *HindIII/SalI* restriction sites and three protection bases were introduced. PT67 (murine (mouse) fibroblast) cells were transfected with phospholipid DOTAP (1,2-dioleoyl-3-trimethylammonium-propane), and HBD-2 gene expression was identified in target cells using Western blotting. RT-PCR revealed that the full-length sequence of the CEA signal peptide was 102 bp, the enterokinase recognition site had a sequence of 15 bp, and the recombinant gene fragment produced by PCR was 136 bp (Zhang et al., 2014).
- *Expression of Magainin2 fusion protein in CHO-s cells:* The design and synthesis of a Magainin2 fusion protein gene that is compatible with mammalian expression systems provide a promising alternative to conventional antibiotics for combating drug-resistant pathogens. The fusion protein gene was designed based on the published amino acid sequence of Magainin2, with an emphasis on using codons biased toward mammalian cells to optimize gene expression in these cells. The resulting gene encodes a 210-amino acid protein composed of the signal peptide of goat beta-casein and the Magainin2 mature peptide. Unexpected splicing events, which can interfere with the level of gene expression, pose a significant obstacle when expressing peptides in mammalian cells. To address this issue, a Northern hybridization analysis was conducted on the transgenic cell line (CHO-s) to identify any cryptic splicing sites, thereby ensuring correct expression in the mammalian system. The development of stable transgenic cell lines and transgenic animals also presents obstacles. In this study, transgenic mice were produced by microinjecting pronuclear embryos, and their genomic DNA was analyzed by PCR and Southern hybridization to confirm the presence of the foreign gene. Notably, the expression level of the fusion protein varied significantly between different founder lines, most likely due to positional effects, gene silencing, and changes in the genetic background during the foreign gene passage from F0 to F1. Researchers can continue to optimize peptide expression in mammalian systems by understanding the difficulties and trade-offs associated with gene design, unexpected splicing events, and the production of transgenic cell lines and

animals. This study demonstrates the viability of using mammalian expression systems for the production of antibacterial peptides such as Magainin2, which could lead to the development of novel therapeutics for the treatment of drug-resistant pathogens (Fan & Li, 2009).

- *Expression of Cecropin A2 in Chinese hamster lung cells (V79)*: Researchers expressed the *Drosophila* antibacterial peptide cecropin A2 in a Chinese hamster lung cell line (V79) to investigate posttranslational modifications and develop potential substitute antibiotics. From plasmid k-15, Cecropin A2 cDNA was extracted, digested with *HindIII* and *XbaI* restriction enzymes, and purified. The cecropin A2 gene-containing recombinant plasmid pRc/RSVcecA2 was introduced into V79 cells through calcium phosphate coprecipitation. Stable clones were obtained after transfection, and genomic Southern blotting confirmed the acquisition of the antibacterial peptide construct in the stable cells. Northern blot analysis evaluated the transcriptional level of expression of the cecA2 gene in stably transfected cells, whereas Western blot analysis employed an antibody raised against *Hyalophora preprocecropin* A (ppCecA) with purified ppCecA as a positive control. To successfully engineer the V79 cell line to express cecropin A2, several obstacles must be overcome, including balancing production costs and therapeutic applications and achieving stable expression while preserving cell line viability. Despite these obstacles, the study detected cecropin gene expression at the peptide level, allowing for a more comprehensive investigation of posttranslational modifications. As it has been demonstrated that cecropin and melittin inhibit the replication of the human immunodeficiency virus, additional research into the expression of antibacterial peptides in mammalian cells could lead to the development of novel therapeutics (Pore & Pal, 2000).

These examples demonstrate the flexibility and potential of recombinant DNA technology for the expression of AMPs in mammalian cells. Researchers have successfully expressed bioactive AMPs using this technology. Not only have these studies contributed to our understanding of AMP function and action mechanisms, but they have also opened up new therapeutic and industrial applications. Nevertheless, it is essential to weigh the potential benefits of AMP expression in mammalian cells against the challenges and trade-offs associated with recombinant DNA technology to ensure the future development and application of these technologies responsibly.

3.2 Synthetic biology

The advancement of synthetic biology techniques has created new opportunities for the design, production, and optimization of AMP expression systems in mammalian cells, which can provide important insights into their function and potential medical applications. This involves the application of standard genetic components, such as promoters, ribosome-binding sites, and terminators, to fine-tune the expression of AMP genes. Moreover, synthetic biology enables the incorporation of posttranslational modifications, such as disulfide bond formation and glycosylation, which are frequently required for the function and stability of AMPs.

- *Expression vectors and promoters*: For optimal AMP expression in mammalian cells, the expression vector and promoter must be carefully selected. For high-level expression of foreign genes, strong, constitutive promoters such as the CMV promoter are commonly used (Johari et al., 2022). Nevertheless, the constitutive expression of AMPs can be toxic to host cells, necessitating the use of inducible promoters that allow for controlled expression (Kallunki et al., 2019). The tetracycline-inducible system is a popular choice for regulating gene expression, as it permits precise control over AMP production while minimizing toxicity to the host cell (Pedone et al., 2019). Researchers have developed strategies for the targeted delivery of AMP-encoding genes to specific cell types or tissues in order to improve the specificity and efficiency of AMP expression in mammalian cells. These approaches include the use of tissue-specific promoters, cell-penetrating peptides, or targeted nanoparticles for gene delivery. While these methods can improve the precision of AMP expression, they may also introduce additional complexity and potential off-target effects, necessitating careful optimization and validation.
- *Signal peptides and secretion*: AMPs must be secreted by the host cell to exert antimicrobial activity. Incorporating a signal peptide sequence into the AMP construct might allow the peptide to be efficiently directed to the secretory pathway and released into the extracellular environment (Dalton & Barton, 2014). The choice of the signal peptide can potentially have a substantial effect on the secretion efficiency and bioactivity of the expressed AMP. For instance, it has been demonstrated that the human interleukin-2 (IL-2) signal peptide increases the secretion of various proteins in mammalian cells (Zhang et al., 2005).
- *Codon optimization and mRNA stability*: The optimization of AMP coding sequences to improve their expression in mammalian cells is likely possible through synthetic biology. Codon optimization, which entails substituting infrequent codons with more common ones, can improve translation efficiency and protein production overall (Mauro & Chappell,

2014). Codon optimization, which entails substituting infrequent codons with more common ones, can improve translation efficiency and protein production overall. In addition, by modifying the untranslated regions (UTRs) of the mRNA, its stability can be increased, thereby potentially enhancing the expression of the AMP (Asrani et al., 2018).

Despite the fact that synthetic biology offers promising avenues for AMP expression in mammalian cells, a number of obstacles and trade-offs must be considered. For example, the potential cytotoxicity of AMPs toward host cells can restrict their therapeutic applications. To address this issue, researchers are investigating methods to optimize AMP sequences and minimize cytotoxicity while maintaining antimicrobial activity (Wu et al., 2014). In addition, the use of synthetic gene circuits and genetically modified cells raises concerns regarding the possibility of off-target effects, unintended consequences, and the ethical implications of such modifications. Another difficulty associated with AMP expression in mammalian cells is the possibility of pathogen resistance. While AMPs are generally believed to be less likely than conventional antibiotics to induce resistance, the possibility still exists, especially with prolonged use or in cases of suboptimal expression. To mitigate this risk, researchers are examining combinatory approaches in which multiple AMPs or AMPs in combination with traditional antibiotics are expressed to enhance the antimicrobial effect and reduce the likelihood of resistance development (Santos et al., 2022).

Examples of protein expression in mammalian cells that could be extended to AMPs using synthetic biology techniques include the following:

- *Synthetic gene circuits*: Synthetic gene circuits are designed to regulate gene expression in response to specific environmental or cellular signals, providing precise control over the production of proteins, including AMPs. Researchers have designed synthetic gene circuits capable of detecting and responding to the presence of a bacterial infection in mammalian cells (Naydich et al., 2019). These types of circuits can be potentially engineered to recognize a particular component of the bacterial cell wall and, through a genetic amplifier, control the expression of an AMP gene. The circuit could be activated upon detecting the presence of bacteria, resulting in the controlled expression and secretion of the AMP. This strategy is likely to not only reduce the cytotoxic effects associated with high levels of AMP expression but also provide a timely and targeted response to bacterial infections.
- *CRISPR-based regulation*: The CRISPR/Cas9 system has revolutionized gene editing and has been repurposed for gene regulation in mammalian cells, providing a potentially versatile platform for the expression of AMPs. Researchers have recently used a catalytically inactive form of Cas9 (dCas9) fused to transcriptional activators to regulate gene expression in mammalian cells (Shakirova et al., 2020). The dCas9-activator complex is recruited to the target site by designing guide RNAs that target the gene promoter, resulting in the upregulation of expression. This strategy can be eventually exploited for precise control of AMP production, minimizing cytotoxic effects and off-target interactions. Moreover, the CRISPR-based system can be easily adapted to regulate multiple AMPs simultaneously, offering a versatile tool for combinatorial therapy against multidrug-resistant pathogens.
- *Directed evolution*: Directed evolution is a potent technique of synthetic biology that could be used to improve the properties of proteins, including AMPs. Researchers used mammalian cell surface display to isolate and engineer a library of antibodies for affinity maturation (Alfaleh, 1986). This methodology can also be utilized to identify AMPs produced by mammalian cells. To generate phage libraries for AMPs expressed in mammalian cells, cDNA libraries from mammalian tissues known to express AMPs, such as skin, mucosal surfaces, and immune cells, could be utilized. By amplifying the cDNA sequences encoding the desired AMPs and incorporating them into phage display vectors, it is possible to generate a phage library specific to AMPs. Alternately, libraries of synthetic peptides can be designed based on known AMP sequences, motifs, or structural characteristics. These synthetic libraries could aid in the identification of novel AMPs with enhanced antimicrobial activity and decreased cytotoxicity. Biopanning can be used to screen phage-displayed AMPs against target bacterial or fungal cells in a manner similar to the selection process for antibody phage display libraries. After multiple biopanning cycles, AMPs with high affinity can be isolated and characterized for their antimicrobial activity, specificity, and potential cytotoxicity to mammalian cells. These peptides can be expressed in mammalian cells to ensure proper folding, posttranslational modifications, and biological activity after the identification of promising AMP candidates. Transient or stable transfection of mammalian cell lines or primary cells with expression vectors encoding the AMP gene of interest can induce AMP expression. After expression, the AMPs can be purified and screened for their antimicrobial activity, stability, and potential for therapeutic use.

Using phage display libraries to discover AMPs expressed in mammalian cells has a number of benefits. The approach can be extended to identify novel AMPs with enhanced antimicrobial activity or reduced cytotoxicity, which could serve as potential therapeutic agents for the treatment of drug-resistant infections, examine the relationship between the structure and activity of AMPs to optimize their activity and reduce potential side effects, discover AMPs with specific activity

against specific pathogens or strains, thereby enabling the development of targeted therapies, and facilitate the development of AMPs as adjunctive therapies to enhance the efficacy of existing antibiotics or reduce the emergence of antibiotic resistance.

An example of enhancing Peptidoglycan Recognition Protein (PGLYRP) expression in HeLa cells through CRISPR/Cas9 Synergistic Activation Mediator (SAM) has been recently published (Bobrovsky et al., 2020). PGLYRPs are innate immune system proteins with antibacterial activity against multiple pathogens, including intracellular bacteria such as *Chlamydia trachomatis*. The objective was to examine PGLYRP's potential as an alternative to conventional antibiotics for fighting drug-resistant bacteria. To accomplish this, the scientists utilized *E. coli* TOP10 bacterial strains and the human cell lines HeLa and HEK 293FT. They utilized the CRISPR/Cas9 SAM system, which consisted of three plasmid vectors, to modify HeLa cells and enhance PGLYRP gene expression. The effectiveness of the infection with chlamydia was determined by counting the number of *C. trachomatis* inclusions in both stable HeLa cell lines producing recombinant PGLYRP and transduced HeLa cells with activated pglyrp gene expression. Contact between PGLYRP proteins and *C. trachomatis* elementary bodies increased the expression of chlamydia's two-component system genes, as demonstrated by the results. This enhanced gene expression has the potential to prevent chlamydial infection. Additionally, HeLa cells with increased PGLYRP gene expression exhibited a decreased susceptibility to chlamydia infection (79.5%–85%). Despite the fact that recombinant proteins offer potential treatment options, they have limitations such as low stability and limited local application possibilities. Gene therapy techniques, such as the CRISPR/Cas9 SAM system, may offer alternative applications for these proteins. However, additional research is required to better comprehend the long-term effects and the possibility of resistance development. The study demonstrates that increasing PGLYRP expression in HeLa cells has the potential to combat chlamydial infections and overcome antibiotic resistance.

3.3 mRNA-based expression systems

In mammalian cells, mRNA-based expression systems have emerged as a potentially versatile and efficient method that can be implemented in the production of AMPs. This technique involves the *in vitro* transcription of mRNA molecules encoding the desired AMP, followed by their delivery to the target cells. In this section, we will examine the various aspects of mRNA-based expression systems in greater detail, including mRNA modifications, delivery methods, and translational regulation.

3.3.1 mRNA modifications

mRNA molecules can be modified in numerous ways to improve their stability and translation efficiency. These include the addition of a 5' cap structure, typically 7-methylguanosine, to the mRNA molecule that enhances its stability and facilitates translation initiation by binding to the eukaryotic translation initiation factor 4E (eIF4E) (Mitchell et al., 2010), adding a polyadenylated tail to the 3' end of the mRNA molecule increases its stability and translation efficiency by preventing exonuclease-mediated degradation and facilitating interactions with poly(A)-binding proteins (Liu et al., 2022), and incorporating chemically modified nucleotides, such as pseudouridine or N1-methylpseudouridine, can decrease mRNA immunogenicity and boost translation efficiency (Nance & Meier, 2021).

3.3.2 mRNA delivery methods

The efficient delivery of mRNA molecules into mammalian cells is a critical factor for successful protein expression and must be considered also for AMPs. Several methods have been developed, including:

1. **Lipid nanoparticles (LNPs):** LNPs are composed of lipids that form a bilayer structure, encapsulating and protecting the mRNA molecule from degradation. LNPs can facilitate endocytosis-mediated uptake of mRNA into mammalian cells and have been used in several mRNA-based AMP expression studies (Hou et al., 2021).
2. **Electroporation:** Electroporation involves the application of an electric field to the cell membrane to create transient pores that allow mRNA molecules to enter the cell. This method has been utilized for the mRNA-based expression of numerous proteins, including AMPs, but it can cause cell damage and low transfection efficiency in certain cell types (Young & Dean, 2015).
3. **Cationic polymers and liposomes:** Cationic polymers, such as polyethyleneimine (PEI), and liposomes can be utilized to form complexes with negatively charged mRNA molecules, thereby facilitating their uptake by mammalian cells. Several protein expression studies have utilized these techniques, but their efficacy can be affected by variables such as polymer-to-mRNA ratio and cell type (Sriwidodo et al., 2022).

3.3.3 Translational regulation and optimization

Optimizing the mRNA sequence and translation initiation is essential for achieving efficient AMP expression. This can be achieved by:

1. **Codon Optimization:** Codon optimization is the process of modifying the coding sequence to match the codon usage bias of the target organism, thereby increasing translation efficiency (Parvathy et al., 2022).
2. **The incorporation of a Kozak sequence,** a conserved sequence surrounding the translation initiation codon, can facilitate translation initiation and increase expression levels (Noderer et al., 2014).
3. **5' and 3' Untranslated Regions (5' and 3' UTRs):** The 5' and 3' UTRs of mRNA molecules can affect translation efficiency and mRNA stability by interacting with various regulatory factors. Optimization of UTR sequences can enhance AMP expression (Asrani et al., 2018).

Despite the fact that mRNA-based expression systems might potentially offer a number of benefits for the expression of AMPs in mammalian cells, there are drawbacks and obstacles to consider when selecting this method. Following is a more in-depth examination of these trade-offs and challenges:

- **Stability and degradation:** mRNA molecules are intrinsically unstable and degradable by cellular ribonucleases. This can result in a shorter half-life for the mRNA and a decrease in protein expression, thereby compromising the efficacy of expression. Among the strategies used to improve mRNA stability are the incorporation of modified nucleotides, which increases the resistance of mRNA to ribonuclease-mediated degradation (Liu & Wang, 2022), the inclusion of stabilizing elements, such as the 5' cap and 3' poly(A) tail, which protect mRNA molecules from exonuclease-mediated degradation and promote interactions with cellular factors that increase mRNA stability (Passmore & Collier, 2022), and the review of 5' and 3' untranslated regions (UTRs), which can affect mRNA stability by interacting with various regulatory factors (Suknuntha et al., 2018).
- **Delivery efficiency:** AMP expression must be robust, for which efficient mRNA delivery into mammalian cells is crucial. Diverse delivery vehicles and techniques have been developed, but their efficacy can be affected by cell type, mRNA size, and the presence of serum proteins. Challenges related to mRNA delivery efficiency include the selection of a suitable delivery vehicle or method, which can vary depending on the targeted cell type or tissue, the optimization of delivery conditions, including the concentration of messenger RNA (mRNA), the presence of serum proteins, and the timing of delivery, to maximize uptake and minimize toxicity, and overcoming extracellular and intracellular barriers to mRNA delivery, such as extracellular ribonuclease degradation, endosomal entrapment, and inefficient release of mRNA into the cytosol (Wadhwa et al., 2020).
- **Immunogenicity:** mRNA molecules can activate pattern recognition receptors such as Toll-like receptors (TLRs) and retinoic acid-inducible gene I (RIG-I)-like receptors (RLRs) to trigger innate immune responses. This can result in the production of inflammatory cytokines, which can reduce the efficacy of mRNA expression and cause undesirable side effects. Strategies to minimize mRNA immunogenicity include incorporating modified nucleotides, which can inhibit TLR and RLR activation (Karikó, 2021), reviewing mRNA sequence optimization, which can reduce the formation of immunostimulatory secondary structures, and being aware that an immunomodulatory agent codelivery can inhibit the activation of innate immune pathways (Nash et al., 2021).
- **Cost and scalability:** The production of mRNA molecules can be costly and difficult to scale, particularly when incorporating modified nucleotides or developing specialized delivery vehicles. Future advances in mRNA synthesis and purification technologies are anticipated to resolve these issues. Key factors that can influence cost and scalability include the cost of raw materials, such as modified nucleotides, which can be significantly more expensive than unmodified nucleotides, the complexity of mRNA synthesis and purification, which can necessitate the use of specialized equipment and knowledge, and the development and production of mRNA delivery vehicles, such as lipid nanoparticles or cationic polymers, can incur additional costs and manufacturing challenges.

The expression of AMPs by mammalian cells is modulated in response to the presence of pathogens, according to recent research. For instance, a recent study investigated the factors that impact the increased mRNA expression of selected AMPs in mammalian cells, such as the beta-defensin (DEFB) family and the S100 calcium-binding protein (S100A) family (Ibrahim et al., 2016). During estrus, when the cervix is open, and bacteria can invade during mating or artificial insemination, the DEFB and S100A protein families may play a role in the immune response. AMP expression by cervix cells is influenced by a variety of factors, including the estrous cycle, uterine health, and bacterial infections. In response to these factors, the DEFB and S100A families exhibit different mRNA expression patterns. mRNA expression of DEFB1 is significantly influenced by the estrous cycle, whereas mRNA expression of S100A11 is unaffected by uterine health status.

The mRNA expression of selected AMPs correlates with polymorphonuclear neutrophil (PMN) content, indicating a connection between the immune response and peptide expression. The complex interactions between the immune system, cellular processes, and external factors such as bacterial infections make it difficult to control the variables that influence AMP expression in mammalian cells. Additionally, different AMP families respond differently to these factors, complicating the development of a universal approach to modulate their expression. Consideration of the potential impact of peptide expression on the immune response, cellular functions, and overall organism health is crucial. Understanding the factors affecting AMP expression and their interplay is crucial for the development of strategies to modulate expression and enhance immune responses against bacterial infections.

Another study demonstrated that mouse B and T cells express and respond to mCRAMP (mouse cathelicidin-related AMP) (Kin et al., 2011). In this study, splenocytes from C57BL/6 mice were sort-purified, with postsort analysis showing greater than 95% purity. B and T cells were found to express Camp mRNA and mCRAMP protein, albeit at lower levels compared with purified neutrophils. The study also investigated the direct effect of mCRAMP on the production of T-cell cytokines, antibody responses, and class switch recombination (CSR). Under conditions that induce Th2, Camp-deficient T cells produced more IL-4, while Camp-deficient B cells produced less IgG1 and IgE antibodies in vitro. The researchers found that Camp-deficient B cells produced less IgG1 mRNA per cell. The study revealed that camp-deficient mice immunized with TNP-OVA/Alum produced greater amounts of TNP-specific IgG1 antibody than wild mice. The immunomodulatory role of AMPs in regulating adaptive immune antibody responses has not been extensively investigated, and existing results are contradictory.

mRNA-based expression systems are a promising method for the expression of AMPs in mammalian cells due to their rapid and transient protein expression, low risk of genomic integration, and the ability to optimize mRNA sequences. Nonetheless, this method presents difficulties in terms of stability, delivery efficiency, immunogenicity, and cost. Researchers can make informed decisions regarding the optimal method for expressing AMPs in mammalian cells by weighing the advantages and disadvantages of mRNA-based expression systems. As mRNA-based expression technologies continue to advance, they have the potential to play a major role in the development of novel antimicrobial therapies, thereby contributing to the ongoing fight against antibiotic resistance and infectious diseases. Future research should concentrate on enhancing the stability of messenger RNA, developing efficient and safe delivery methods, minimizing immunogenicity, and reducing production costs. By overcoming these obstacles, mRNA-based expression systems can provide a potent tool for the expression of AMPs in mammalian cells, opening the door to new therapeutic applications for the treatment of infectious diseases and beyond.

This section has provided a comprehensive analysis of the three most important techniques for AMP expression in mammalian cells: recombinant DNA technology, synthetic biology, and mRNA-based expression systems. Each of these methods offers distinctive benefits, drawbacks, and difficulties that must be carefully considered when determining the most suitable approach for a given application (Table 8.3).

Recombinant DNA technology is the most established method, offering a simple and cost-effective way to produce AMPs. However, it is constrained by potential toxicity, complex purification procedures, and the risk of horizontal gene

TABLE 8.3 Comparison of technologies for the expression of AMPs in mammalian cells.

Approach	Advantages	Trade-offs	Challenges
Recombinant DNA technology	Stable and long-term expression Established methods and protocols High protein expression levels possible	Risk of genomic integration Potential insertional mutagenesis	Cloning and selection process Potential silencing of transgenes Ethical concerns related to genetic manipulation
Synthetic biology	Precise control of gene expression Programmable and tunable systems Potential for dynamic and responsive expression	Complexity of genetic circuit design Potential off-target effects	Design and assembly of genetic circuits Design and assembly of genetic circuits Integration of circuits into host genome or vector
mRNA-based expression systems	Rapid and transient protein expression No risk of genomic integration Ability to optimize mRNA sequences and modifications	Limited mRNA stability Delivery efficiency Immunogenicity	mRNA stability and degradation Efficient delivery methods and vehicles Immunogenicity and immune response modulation Production cost and scalability of mRNA and delivery vehicles

transfer. Synthetic biology, on the other hand, enables the design of AMPs with enhanced properties, but it is still an emerging field that may necessitate significant time, labor, and resource investments. Finally, mRNA-based expression systems offer a promising alternative for rapid, transient, and localized AMP production without the risks of genomic integration; however, obstacles such as mRNA instability, delivery, and immunogenicity must be addressed.

Future research should address the limitations and optimize the efficacy of each approach, as well as investigate their potential synergies. In addition, advancements in bioinformatics, high-throughput screening, and genome editing technologies are anticipated to play a significant role in enhancing the design and expression of AMPs in mammalian cells. Overall, the continued development of these expression systems is crucial for maximizing the potential of AMPs as a viable and effective alternative to conventional antibiotics in the ongoing fight against multidrug-resistant pathogens.

4 Exploring the drivers and challenges of biological antimicrobial production in plants, insects, and mammalian cell systems

4.1 Motives for utilizing plant, insect, and mammalian cell systems for the production of AMPs

In this section, we will discuss the importance of using plant, insect, and mammalian cell systems as biological platforms for the production of AMPs.

- *Driver 1: Proper folding and posttranslational modifications:* Plant, insect, and mammalian cell systems offer the advantage of proper folding and posttranslational modifications, which are crucial for the bioactivity, stability, and solubility of AMPs. These systems can mimic native folding patterns and perform complex modifications such as glycosylation and disulfide bond formation, which may not be achievable in bacterial and yeast systems (Gómez et al., 2014).
- *Driver 2: Reduced toxicity concerns:* The use of plant, insect, and mammalian cell systems can reduce the concerns related to the potential toxicity of AMPs to the host organism used for production. These systems are less sensitive to the cytotoxic effects of AMPs, allowing for efficient production without compromising cell growth or viability (Sampaio de Oliveira et al., 2020).
- *Driver 3: Ability to produce large and complex peptides:* Plant, insect, and mammalian cell systems can efficiently produce large and complex peptides that may be challenging to express in bacterial and yeast systems due to their limitations in processing and folding (Georgianna and Mayfield, 2012).
- *Driver 4: Scalability and sustainability:* In the case of plant-based production, the use of transgenic plants for AMP production provides a scalable and sustainable platform, as plants can be grown in large quantities using renewable resources, reducing the environmental impact of production (Schillberg & Finnern, 2021).

4.2 Challenges

- *Regulatory and biosafety concerns:* The use of plant, insect, and mammalian cell systems for AMP production involves regulatory and biosafety concerns. The release of transgenic plants or genetically modified insects into the environment raises potential risks, such as gene flow, nontarget effects, and adverse ecological consequences (Hilbeck et al., 2011). Therefore, extensive risk assessment and regulatory approval are necessary before these systems can be widely used for AMP production (Yan and Kerr, 2002).
- *Higher production costs:* Compared with bacterial and yeast systems, plant, insect, and mammalian cell systems can have higher production costs, mainly due to their complex growth requirements, lengthy production times, and the need for specialized equipment and facilities. To overcome these challenges, strategies such as optimizing growth conditions, improving expression vectors, and using cost-effective media formulations can be employed (Swiech et al., 2012).
- *Protein expression and purification:* Another challenge associated with plant, insect, and mammalian cell systems is the low expression levels and difficulties in purification. The use of signal peptides, optimizing gene constructs, and employing fusion tags can help improve expression levels and facilitate purification (Gecchele et al., 2015).

5 Future perspectives

Despite their potential, producing AMPs on a large scale remains a daunting challenge. Traditional methods for producing AMPs—chemical synthesis and recombinant expression in bacteria, yeast, fungi, insect cells, mammalian cells, and plants—have several limitations, such as low yields, high costs, and complex purification processes. As such, there has

been an increasing interest in exploring novel biologically based expression systems for AMPs that can circumvent these issues. Other than novel expression systems, there are other important factors that must be considered to ensure the development and successful application of these antimicrobial agents. These include optimizing AMP production, protein folding, and posttranslational modifications, as well as safety evaluation and in vivo activity. They also help overcome regulatory and commercialization hurdles. These multidimensional issues are essential to advancing AMP research and development. They also contribute to the global effort to fight antimicrobial resistance and open the door to innovative and effective treatments for infectious diseases.

5.1 Novel expression systems for AMPs

In this section, we discuss three potentially novel biologically based expression systems for AMPs.

5.1.1 *Microalgae*

Microalgae are photosynthetic microorganisms that have recently gained attention as potential expression systems for recombinant proteins (Malla et al., 2021). Microalgae possess several advantages as a production platform, including rapid growth rates, the ability to utilize light energy and carbon dioxide, and the potential for large-scale cultivation in outdoor systems. Additionally, microalgae are generally recognized as safe (GRAS) and are not known to harbor human pathogens, reducing the risk of contamination in the final product (Wong et al., 2022). However, microalgae expression systems also present several challenges. First, the development of efficient transformation and selection methods is still in its infancy (Grama et al., 2022). Second, the downstream processing of AMPs from microalgae may be complex due to the presence of various cell wall components and other intracellular components (Rahman et al., 2022). Nonetheless, with continued research and optimization, microalgae could become a promising platform for the production of AMPs.

5.1.2 *Protozoa*

Protozoa are unicellular eukaryotic organisms that have not been extensively explored as expression systems for recombinant proteins. However, they have several characteristics that make them a potential candidate for AMP production. Protozoa can grow rapidly, have simple nutritional requirements, and can be cultivated in suspension, facilitating large-scale production (Niimi, 2011). One of the primary challenges of using protozoa as an expression system is the lack of established genetic engineering tools and protocols. Moreover, some protozoa are known to be human pathogens, raising concerns about the safety of using them as expression hosts. Further research is required to identify suitable protozoan species for AMP production, develop genetic engineering techniques, and address safety concerns.

5.1.3 *Bacteriophages*

Bacteriophages, commonly referred to as “phages,” are viruses that infect bacteria. Phages have been extensively researched for their potential use in phage therapy—a method to fight bacterial infections (Lin et al., 2017). Recently, there has been increased interest in exploring the potential of phages as expression systems for AMPs. Phages can be engineered to display AMPs on their surface or produce them within bacteria upon infection (Davydova, 2022). Phage expression systems for AMPs offer several advantages, such as high specificity toward target bacteria, self-replication, and rapid evolution. However, this approach also comes with its share of challenges. For instance, the expression of AMPs within bacteria may lead to cell lysis, decreasing production yields (Egido et al., 2023). Furthermore, the regulatory framework for using bacteriophages in therapeutic applications is still being developed and could potentially restrict the adoption of this technology (Fauconnier, 2019).

5.1.4 *Trade-offs and challenges*

In developing novel biologically based expression systems for AMPs, it is crucial to consider the trade-offs involved in balancing different factors. For instance, while some expression systems may offer high production yields, they may also be associated with complex downstream processing or safety concerns. Similarly, the use of photosynthetic organisms such as microalgae may reduce production costs by utilizing sunlight and CO₂, but the development of efficient genetic engineering tools remains a challenge. Furthermore, each expression system comes with its own set of challenges. For microalgae, optimization of transformation and selection methods is required, while protozoa demand the development of genetic engineering tools and protocols as well as addressing safety concerns. In the case of bacteriophages, balancing the production yield with potential host cell lysis and navigating regulatory frameworks are key challenges.

5.1.5 *Environmental and ethical considerations*

When making decisions regarding novel biologically based expression systems for AMPs, it is essential to take into account both the environmental and ethical impacts of each approach. For instance, using microalgae as an expression system could contribute to carbon sequestration and reduce greenhouse gas emissions; however, large-scale cultivation in open systems could contaminate natural water bodies, potentially impacting local ecosystems ([Wimmerova et al., 2023](#)).

Similar to protozoa or bacteriophages as expression systems, using them for expression may raise ethical issues if they involve the release of genetically modified organisms into the environment. Therefore, it is essential to develop containment strategies and conduct thorough risk assessments to minimize potential negative impacts on ecosystems and human health.

5.2 *In vivo activity and safety*

It is crucial to evaluate the *in vivo* activity of AMPs to develop effective antimicrobial treatments. *In vitro* tests, such as the determination of the minimum inhibitory dose (MIC) or time-kill assays, can provide preliminary insights into the antimicrobial properties of AMPs. These assays cannot predict *in vivo* efficacy because they don't account for factors such as pharmacokinetics or protein stability. For *in vivo* testing of AMPs, scientists can use a variety of animal models, such as murine, zebrafish, or insect models. These models allow for the evaluation of AMPs' efficacy in clearing the target organism, their impact on host life, and potential synergistic or antagonistic interactions. Additional *in vivo* imaging methods, such as fluorescence imaging or bioluminescence imaging, provide real-time insights into infection dynamics and the distribution of AMPs in the host.

It is essential to ensure the safety of AMPs used for therapeutic purposes. This involves evaluating their potential to be immunogenic and toxic and inducing off-target effects in relevant animal models and then, ultimately, in human clinical trials. It is possible to assess the potential immunogenicity of AMPs by monitoring the host's immune response to the peptides. This includes the production of anti-peptide antibodies and the activation of immune cells. Toxicity assessment can include assessing the effect of AMPs on host cell viability, organ function, and overall health and determining the Therapeutic Index, which is the ratio between the toxic dose and the effective dose. Research can help ensure that effective, safe, and clinically proven antimicrobial therapies are developed by thoroughly examining the *in vivo* activity of AMPs.

6 Conclusion

The rising prevalence of antibiotic resistance has highlighted the urgent need to find novel antimicrobial strategies. These include the development and optimization of biologically based expression systems for AMPs. We have examined the potential of microalgae/protozoa and bacteriophages for the production of AMPs. They are a promising alternative to traditional expression systems such as bacteria, yeast, fungi, or insect cells. Each of these innovative expression systems provides unique advantages and potential for the production of AMPs. Because of their rapid growth, ability to harness carbon dioxide and light energy, and compatibility with large-scale outdoor cultivation, microalgae could be a cost-effective and long-lasting option. While relatively new, protozoa have the potential to grow quickly and require very few nutritional requirements. They can also be easily grown in suspension culture.

Bacteriophages provide high specificity, self-repetition, and the potential for rapid evolution, which makes them attractive candidates for AMP manufacturing. The downside is that embracing novel expression systems can present its own challenges and trade-offs. To make microalgae more productive, genetic engineering tools are needed. The identification of appropriate protozoan species and safety concerns are critical in the case of protozoa. This is why genetic engineering techniques have been developed. Issues facing bacteriophages are production yields and managing the regulatory landscape of phage-based therapeutic applications. It is important to think about the ethical and environmental impacts of these new expression systems. Microalgae could, for instance, help reduce greenhouse gas emissions by storing carbon, while large-scale cultivation could negatively impact local ecosystems.

In addition, genetically modified protozoa and bacteriophages can pose ethical issues and require containment strategies to reduce their potential adverse effects on ecosystems and human health. Continued research and interdisciplinarity are crucial to overcoming these difficulties and unlocking the full potential of these innovative expression systems. If we invest in the development of protozoa and microalgae-based platforms, it will be possible to speed up the production of new antimicrobial strategies to counter the increasing threat of antibiotic resistance. By doing so, we can improve global health outcomes and promote a more sustainable, responsible approach to biopharmaceutical product production. Researchers must also explore new expression systems. They must address key aspects such as optimizing AMP production, protein

folding, and posttranslational modification, as well as ensuring safety and in vivo activity. Optimizing AMP manufacturing requires the selection and design of appropriate expression hosts.

Researchers can enhance the therapeutic power of these promising antimicrobial agents by maximizing their yield and quality. Protein folding and posttranslational modifications are crucial in determining the bioactivity as well as the stability and functionality of AMPs. For AMP-based therapies that are effective, it is vital to ensure proper folding and modifications of the expression host. Researchers have many options to tackle these issues, including coexpression with chaperone proteins, genetic engineering in host organisms, and the use of cell-free synthesis systems. For the development of safe and effective antimicrobial medicines, it is essential to evaluate the safety of AMPs and their in vivo activity. To ensure that in vivo safety is maintained, researchers must use suitable animal models for infection. These important factors, along with the development of novel expression platforms, can all be used by researchers to contribute to the success of antimicrobial therapies. This holistic approach to AMP development is crucial for the ongoing battle against antibiotic resistance and the pursuit of innovative solutions to the global healthcare crisis.

7 AI disclosure

During the preparation of this work the author(s) used GPT 4 as style and grammar checker throughout the chapter. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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ANTIMICROBIAL PEPTIDES

A Roadmap for Accelerating Discovery and Development

Edited by

Luis H. Reyes, Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia

Juan C. Cruz, Department of Biomedical Engineering, Universidad de los Andes, Bogotá, Colombia

Gregory R. Wiedman, Department of Chemistry and Biochemistry Seton Hall University, South Orange, NJ, United States

Contains a comprehensive description of computational and experimental approaches for the discovery, manufacture, and clinical translation of antimicrobial peptides.

Antimicrobial Peptides: A Roadmap for Accelerating Discovery and Development covers the most important efforts of scientists and engineers worldwide to accelerate the process of discovery, production, and eventual market penetration of more potent antimicrobial peptides. These efforts have been fueled by emerging technologies such as artificial intelligence and data science, molecular and CFD simulations, easy-to-use process simulation packages, microfluidics, and 3D printing, among many others. Such technologies can now be implemented and scaled up quickly and at relatively low cost in low-budget production facilities, critical to moving to sustainable and marketable products worldwide.

Discovering novel antimicrobial peptides rationally and cost-effectively has emerged as one of the significant challenges of modern biotechnology. Thus far, this process has been tedious and costly, resulting in molecules with activities far below those needed to address the current challenge of microbial resistance to antibiotics that takes the lives of thousands of people around the world every year. Due to these issues, large biopharmaceutical companies have decided not to invest in discovering novel antimicrobials, which is worrisome in the long run due to the possibility of a new pandemic.

This book also delineates how multidisciplinary teams have assembled to address the challenges of manufacturing, biological testing, and clinical trials to finally reach complete translation. It is an ideal resource for scientists, researchers, engineers, and students in the fields of biotechnology, microbiology, biochemistry, and pharmaceuticals. Finally, anyone interested in the emerging technologies that are enabling the discovery and development of new antimicrobial peptides may find this book interesting and informative.

Key Features

- Covers computational tools (including emerging artificial intelligence algorithms) and microfluidic systems for discovering and high-throughput screening of AMPs
- Discusses the application of bioprocess engineering scale-up approaches for AMPs production and purification with the aid of process simulation tools and rapid prototyping
- Highlights user-centered design and formulation of products with AMPs
- Describes the whole pipeline for AMPs production



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